

Smart Attend System

Tools & Technologies Used — Backend

1. Runtime & Language

Tool	Version	Usage
Node.js	v18	Runs JavaScript on the server
JavaScript	ES Modules	Project language — uses import/export

2. Framework

Tool	Version	Usage
Express.js	v5	Builds API routes and handles HTTP requests

3. Database

Tool	Version	Usage
MongoDB	Latest	NoSQL database — stores all project data
Mongoose	v8	ODM — defines schemas and talks to MongoDB

4. Authentication & Security

Tool	Version	Usage
jsonwebtoken (JWT)	v9	Generates and verifies auth tokens
bcrypt	v6	Hashes and compares passwords
cors	v2	Allows mobile app and web to call the API

5. Validation

Tool	Version	Usage
Joi	v18	Validates request body data (signup, login)

6. Email Service

Tool	Version	Usage
Resend	v6	Sends OTP and notification emails via API (Railway blocks SMTP)

7. QR Code

Tool	Version	Usage
qrcode	v1.5	Generates daily QR code image for each employee

8. PDF Generation

Tool	Version	Usage
pdfkit	v0.17	Creates attendance report PDF files

9. Hosting & Deployment

Tool	Usage
Railway	Hosts the Node.js server and MongoDB database
GitHub	Version control and auto-deploy to Railway on push

10. Testing

Tool	Usage
Postman	Tests all API endpoints manually

Quick Summary

Category	Tool
Runtime	Node.js v18
Framework	Express.js v5
Database	MongoDB + Mongoose
Auth	JWT + bcrypt
Validation	Joi
Email	Resend
QR Code	qrcode
PDF	pdfkit
CORS	cors
Hosting	Railway
Version Control	Git + GitHub
Testing	Postman